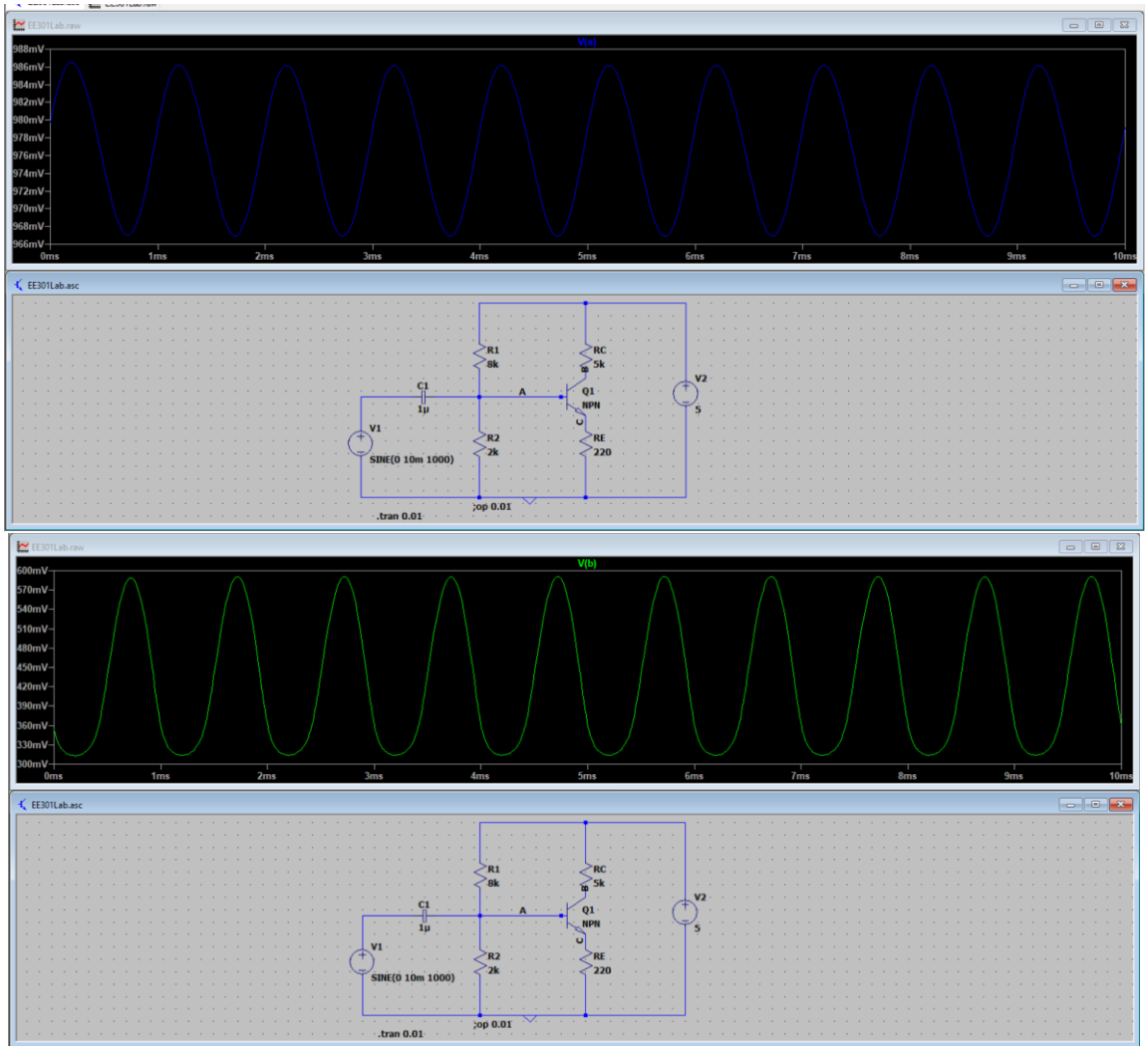
1. Introduction

In this lab, we look into the practical aspects of Bipolar Junction Transistors (BJTs) and MOS transistors, aiming to gain insights into transistor characteristics. Our focus centers on measuring current gain and obtaining hands-on experience with transistor circuits. This exploration is crucial for understanding the fundamental behavior and applications of these electronic components

2. Preliminary Work

2.1. In the following circuit Cir 2.2, we tried to calculate R1, R2, R_C and R_E to achieve gain of 10 and noted the following outputs:

- Choose R1 and R2 such that the design is independent of Beta.
- Choose RE such that the design is not prone to VBE variations.
- Choose RC to get gain of 10.



Gain is 300/20 = 15

We can calculate gain to 15 with

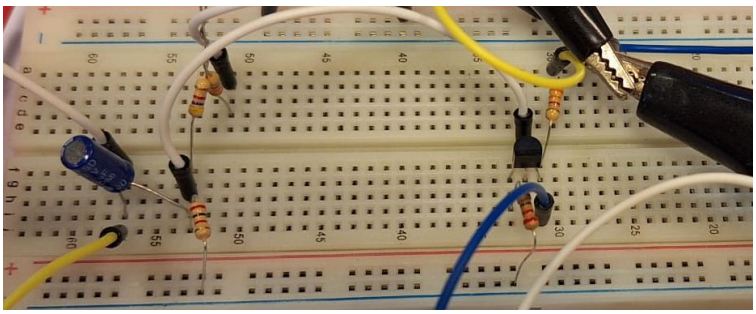
R1 = 8k
R2 = 4k
R_C = 5k
R_E = 220

With increasing value of R_E we could get more precise results, but I have learned that after the laboratory session. So I have just moved with gain 15.

3. Lab Tasks

Task 1:

Constructed circuit 2.2 with values found in preliminary work.



And measured the V_{OUT} while increasing the Frequency from 100 Hz to 100kHz.

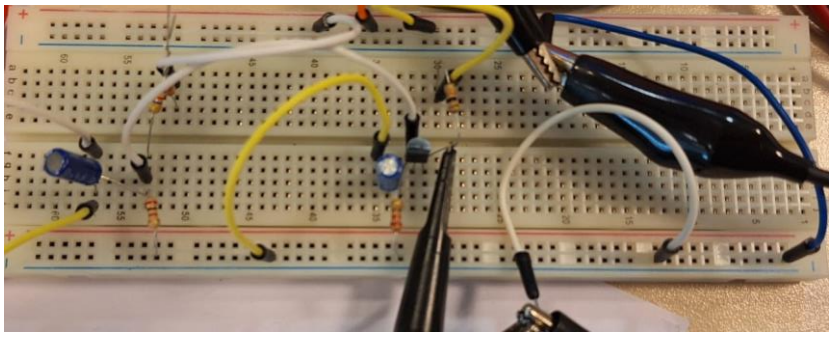
Freq	Vout
100	170
200	190
300	200
500	210
600	212
700	215
800	220
1k	215
3k	218
5k	245
10k	215
30k	216
60k	214
70k	220
80k	225
90k	216
100k	210

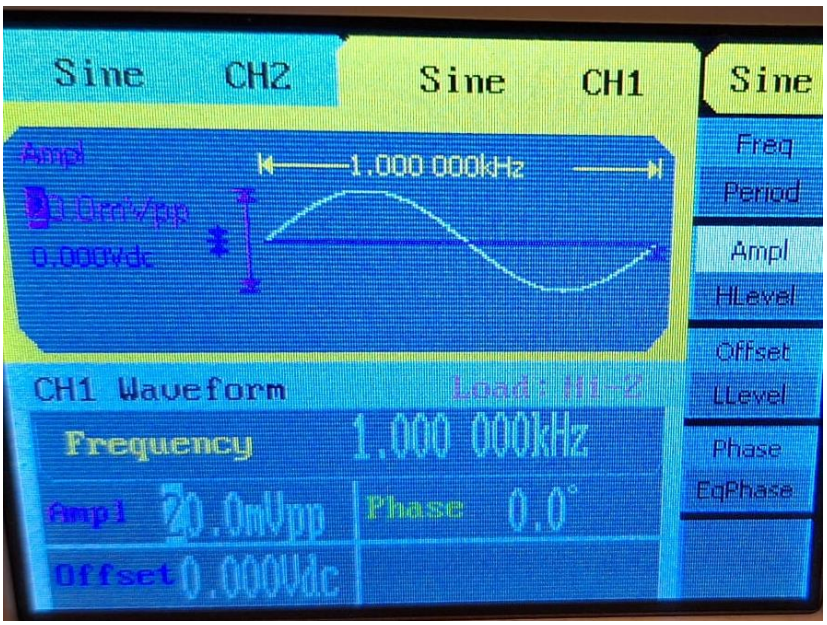
- Around 230mVpp my sinus wave started to break.

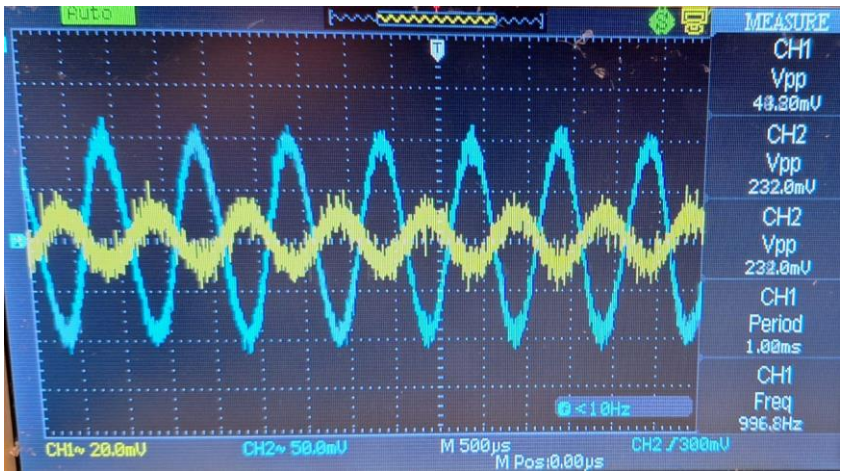
When I changed my transistor with another unused transistor, output decreased because that transistor did not used as much as I used that. When I used my transistor constantly it heated up so we get different values from the measurements.

Task 2:

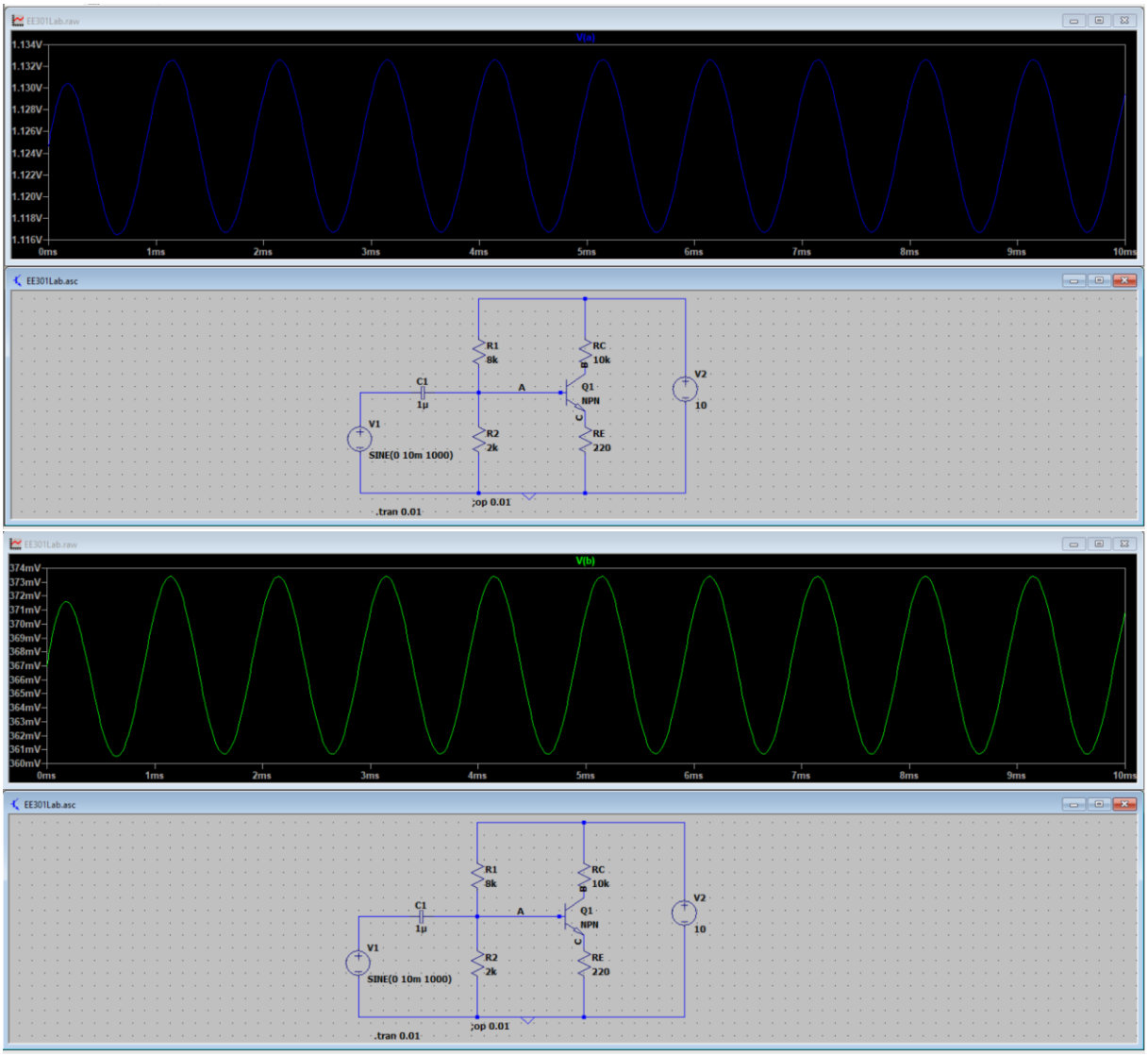
Constructed the circuit shown in Cir 2.5 with 2N7000 NMOS transistor with applying a sinusoidal voltage (20 mVpp at 1 kHz). We changed 1k resistor to 10k and applied 10V instead of 5V.







We could not get desired results from the circuit in real life but on the LTSpice results were satisfying.



Task 3:

Task 3 left blank because of the time issues.

Conclusion

In summary, our exploration of Bipolar Junction Transistors (BJTs) and MOS transistors provided practical insights into current gain measurement and hands-on experience with transistor circuits. The preliminary design achieved a gain of 15, but further precision is possible with extended experimentation.

Sadly we could not finished experiment.